Floor Conn

J6
Micro-Fit 3.0 2x6 (43045-1213)

+24V, GND, SD_X, SC_X, nRESET_X, OE_X
SDA, SCL, nRESET, OE
GND, 1-WIRE, +5V
1-Wire_X, +5V

+5V C21 10uF GND
+5V C59 100nF GND

D45 ESD5Z3.3T1G GND

Floor ID

U18
DS28E07Q_T

1-Wire_X — IO, EP, GND
D29 ESD5Z6.0T1G

ADC Module

U16
ADS1115IDGSR

AIN0, AIN1, AIN2, AIN3
VDD, ALERT/RDY, SDA, SCL, ADDR, GND
+3V3_SENS
C67 1uF GND
C68 100nF GND
SDA_S, SCL_S

295 1M: at the electrode (not on the PCB)
TRAY_B — R95 1M — TRAY_A

J7
Tray electrode (JST PH 2p)
TRAY_B, TRAY_A
D25 ESD5Z3.3T1G GND
R101 100K GND
C64 10nF GND
R103 1K
D24 ESD5Z3.3T1G GND
R99 10K TRAY_EXC

Spectral Sensor

J20 = Adafruit 4698 (AS7341 + 1.8 V LDO + level shifter)
1 VIN 2 GND 3 SCL 4 SDA 5 GPIO 6 INT

J20
+3V3_SENS, GND, SCL_S, SDA_S

AS7341 breakout (Adafruit 4698)

MECHANICAL

H1 M3, H2 M3, H3 M3, H4 M3
FID1 Fiducial, FID2 Fiducial, FID3 Fiducial
Hole positions will be assigned by the partner

LED Driver

U19
PCA9685BS_118

+3V3
R105 10K
OE_X — D26 ESD5Z3.3T1G
SCL_X, SDA_X
A0, A1, A2, A3, A4, A5, EXTCLK

+3V3
C71 1uF
C72 100nF
VDD, EXP, VSS
LED0 CH1, LED1 CH2, LED2 CH3, LED3 CH4, LED4 FAN_PWM
LED5–LED14 CH3/CH4 are RESERVED
LED15 TRAY_EXC

Bus Buffer

U15A 74AHCT125
CH1 — R127 100K — CH1_G, OE_X

U15C 74AHCT125
CH3 — R128 100K — CH3_G, OE_X

U15B 74AHCT125
CH2 — R129 100K — CH2_G, OE_X

U15D 74AHCT125
CH4 — R130 100K — CH4_G, OE_X

+5V
C85 100nF
U15E 74AHCT125
VCC, GND

Usable LED strips

+24V
C73 220uF, C75 1uF, C77 100nF
D30 SS25 W_LED-
CH1_G — R111 33 — Q14 DMN4035L-7 — R113 100K
C74 220uF, C76 1uF, C78 100nF
D31 SS25 R_LED-
CH2_G — R112 33 — Q15 DMN4035L-7 — R114 100K
GND

CH3/CH4: laid out, not fitted (DNP)

+24V
D32 SS25 CH3_LED-
CH3_G — R115 33 — Q16 DMN4035L-7 — R117 100K
D33 SS25 CH4_LED-
CH4_G — R116 33 — Q17 DMN4035L-7 — R118 100K

FAN OPTION (v2, all DNP): local 24 V -> 12 V buck; the fan switch and J19/J21 are in the "FAN SWITCH" block left of the I2C buffer

Floor I2C Buffer

R107/R108 (0R): plain-I2C build only — DNP while U20 is fitted

R107 0
+5V, +5V
C70 100nF GND
+3V3, +3V3
R104 2.2K, R106 2.2K
R109 4.7K, R110 4.7K
SD_X — ESD5Z6.0T1G D27 GND
SC_X — D28 ESD5Z6.0T1G GND
U20
P82B96D
VCC, TX, RX, SX, TY, RY, SY, GND
SDA_X, SCL_X
GND
R108 0

U13 SHT4XI-DIGITAL

SCL_X, SDA_X
SCL, SDA, VDD, VSS
+3V3
C58 100nF GND

Q12 / Q13 BSS138BK-215

+3V3_SENS — Q12 — SDA_X, SDA_S
+3V3_SENS — R125 — Q13 — SCL_X, SCL_S
+3V3_SENS — R126 10K — SCL_S

J3 Light plate (Micro-Fit 1x5)

+24V
W_LED-, R_LED-, CH3_LED-, CH4_LED-

Floor LDO

Might not be needed bead

Probe-Rail LDO
U14
TPS7A2033PDBVR
5V_SENS — C55 1uF GND
IN, EN, OUT, GND, NC
+3V3_SENS
C60 1uF GND
R96 47K +3V3
nRESET_X — C56 10nF GND

Logic LDO
U17
TPS7A2033PDBVR
+5V — C65 1uF GND
+3V3
C69 1uF GND

+5V — FB1 Ferrite 0603 (or 0R) — 5V_SENS
PWR_FLAG: 5V_SENS, +5V, +24V, +12V, GND

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